

Elastomer-Interfaced SMA Hybrid Composites: Interface Enhancement, Analytical Modelling, and Manufacturability for Morphing Laminates

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Abstract

Shape Memory Alloy Hybrid Composites with an elastomeric interface (E-SMAHCs) represent promising candidates for morphing structures. Our previous work demonstrated that a thin elastomeric interlayer around superelastic NiTi wires increases debonding forces and prevents abrupt failure, thereby enabling the full exploitation of the martensitic transformation. In this work, we extended this concept from material characterisation to a functional actuator driven by the Shape Memory Effect (SME). First, we extended pull-out testing to SME wires in martensitic and Joule-heated austenitic states, both with and without pre-strain, confirming that the elastomeric interface can sustain the load throughout the full phase transformation up to wire fracture. Second, to facilitate real-time control without complex models, we formulated an analytical model that approximates the laminate as a Timoshenko bimetallic beam and lumps the SMA-elastomer sub-laminate into a temperature-dependent phenomenological coefficient of thermal expansion (CTE). We then manufactured the laminates using a robust two-step curing strategy to address wire handling, pre-strain retention and embedding of temperature sensors. Finally, we tested the E-SMAHC in a climatic chamber, measuring the curvature, to fit the phenomenological CTE using a sigmoidal function. Our model predicted cantilever tip positions with a root-mean-square error of 6.5% of full scale over the heating branch.

Keywords: *Shape Memory Alloy Hybrid Composites (SMAHC); Elastomeric interface; Morphing structures; Analytical modelling; Experimental Validation.*

1 Introduction

Shape Memory Alloy Hybrid Composites (SMAHCs) combine the large recoverable strains of SMA with the stiffness and low weight of fibre-reinforced polymers. Heating an embedded SMA wire above its austenite start (A_s) temperature triggers a phase transformation, producing macroscopic strains of 3-5%. When the wire is bonded asymmetrically within a laminate, this contraction generates an internal bending moment that morphs the structure [1].

Interface damage and its prevention are one of the main engineering challenges in SMAHC [1]. The large strain mismatch between the deforming NiTi wire and the stiff polymer matrix concentrates shear stresses at the wire-matrix boundary. Because brittle epoxy matrices tolerate minimal shear strain, direct embedding of SMA wires often results in early debonding. In our previous work, we addressed this challenge with an innovative solution: inserting an elastomeric interlayer between the NiTi wire and the composite matrix [2]. Pull-out (PO) tests on superelastic (SE) NiTi wires showed that epoxy-interfaced specimens failed abruptly by brittle delamination, while elastomer-interfaced specimens sustained load through the full stress-induced martensitic (SIM) transformation plateau. Finite element analysis (FEA) with cohesive

zone modelling (CZM) revealed the debonding mechanisms, specifically the redistribution of interfacial shear stresses and the accommodation of the radial contraction of the wire during martensitic transformation. Moreover, residual thermal stress analysis showed that the elastomeric interlayer reduces curing-induced stresses by orders of magnitude.

Despite these advances, two critical gaps remain. First, the Shape Memory Effect (SME) conditions were not tested in PO specimens. Second, no analytical model was developed to predict laminate deflection solely from temperature, which is the minimum requirement for any future closed-loop control. In this paper, we address both gaps through three steps:

- i) extension of PO testing to SME NiTi wires at two embedding lengths, in martensitic phase (*M*) at room temperature and austenitic phase (*A*) by Joule heating, and with or without pre-strain;
- ii) approximation of the E-SMAHC as a Timoshenko bimetallic beam, with a temperature-dependent phenomenological CTE that lumps the phase-transformation contraction;
- iii) manufacturing of an E-SMAHC with embedded temperature sensors using a novel two-step curing strategy and testing in a climatic chamber of the laminate for identification of the phenomenological CTE, fitting and comparison between measures and analytical model results.

2 Materials and Methods

2.1 Materials and Characterisation

We used a NiTi SME wire (SmartFlex 02, SAES Getters) with a 0.2 mm diameter and an austenite start temperature (*A_s*) of 75 °C. We verified that the manufacturer's training was performed via Joule heating at constant stress, yielding a consistent stroke across cycles and full actuation with a 0.4 A DC (used for Joule heating in all subsequent tests). We measured the transformation temperatures by ramping the temperature in a climatic chamber while monitoring displacement under constant stress with 0.5 kg (156 MPa) and 1.9 kg (593 MPa), which, per the datasheet, correspond to 3% and 5% strain, respectively. Subsequently, we tested the wire in uniaxial tension according to ASTM F2516-18 at room temperature (*M*) and under Joule heating (*A*), fitting two multilinear material models for our finite element analysis (FEA).

The elastomeric interface (KRAIBON®) and the GFRP were identical to those used in our previous study, with ply thicknesses of 0.5 mm and 0.22 mm [2]. We co-cured all specimens in an autoclave at 120 °C for 120 min, applying 3 bar of pressure and 0.95 bar of vacuum, with 2 °C/min ramps. We re-characterised the new KRAIBON® batch via tensile and compressive tests and fitted a 5th-degree Mooney–Rivlin model, as in [2]. Furthermore, we performed dynamic mechanical analysis (DMA) on the cured KRAIBON® using a rheometer (HR 20, TA Instruments) in tension mode (10 mm gauge length, 50 µm oscillation, 1 Hz, 2 °C/min temperature ramp).

2.2 Pull-Out Experiments

We manufactured PO specimens (four per condition) by stacking GFRP prepreg and KRAIBON® plies in 3D-printed fixtures as in [2]. We chose two embedding lengths: 1/8", the critical length below which phase transformation does not occur, and 1", which approaches the length at which the wire should break before delamination [2]. For each length, we ran four conditions: room temperature (*M*) or Joule heating (*A*), each with no pre-strain or 5% pre-strain (593 MPa). For the FEA, we applied the 2D axisymmetric CZM model from [2], updating only the hyperelastic model for the KRAIBON® and the multilinear hardening models for the SME wire in both phases.

2.3 E-SMAHC Analytical Modelling, Manufacturing and Testing

We idealised the E-SMAHC laminate as a two-layer beam following Timoshenko's bimetallic theory [3]. The curvature ($\kappa = 1/\rho$) of the uniformly heated beam is given by [3]:

$$k = (\alpha_2 - \alpha_1) \cdot (T - T_0) \cdot \left[\frac{h}{2} + 2 \cdot \frac{(E_1 \cdot I_1 + E_2 \cdot I_2)}{h} \cdot \left(\frac{1}{E_1 \cdot a_1} + \frac{1}{E_2 \cdot a_2} \right) \right]^{-1} \quad (1)$$

We grouped the SMA wires and the KRAIBON[®] elastomer in layer 1. We approximated its elastic modulus using only KRAIBON[®], since the SME wires contribute negligibly to bending stiffness and their axial contribution is lumped into the CTE coefficient. Layer 2 consisted of the GFRP composite. Due to the extreme stiffness contrast between the two layers in the E-SMAHC ($E_1 I_1 / E_2 I_2 \approx 1/153$ and $E_2 a_2 / E_1 a_1 \approx 794$), the denominator is dominated by the GFRP bending stiffness and the KRAIBON[®] axial compliance. Thus, we simplified the equation to:

$$k = (\alpha_2 - \alpha_1(T)) \cdot (T - T_0) \cdot \left[\frac{h}{2} + \frac{2 \cdot E_2 \cdot I_2}{h \cdot E_1 \cdot a_1} \right]^{-1} \quad (2)$$

with $\alpha_2 = 14 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$, $T_0 = 25 \text{ } ^\circ\text{C}$, $h = 1.44 \text{ mm}$, $E_2 = 16.7 \text{ GPa}$, $I_2 = 0.156 \text{ mm}^4$, $E_1 = 8.9 \text{ MPa}$, and $a_1 = 1.0 \text{ mm}$. We define $\alpha_1(T)$ as a positive parameter that models the SMA contraction, so that increasing α_1 produces increasing curvature toward the SMA layer. Because the curvature is driven by crystallographic phase transformation rather than pure thermal expansion, standard micromechanical mixture rules do not apply [4]. Instead, we treated $\alpha_1(T)$ as a phenomenological, phenomenological CTE calibrated to reproduce the measured curvature. To identify this parameter, we heated and cooled the E-SMAHC cantilever in a climatic chamber while optically tracking its shape. We then fitted a sigmoidal function to each thermal branch:

$$\alpha_1(T) = a + \frac{b}{1 + \exp[-m \cdot (T - T_{mid})]} \quad (3)$$

where a is the lower asymptote, b is the transformation amplitude, T_{mid} is the inflection temperature, and m is the steepness. We assumed the cooling value of the lower asymptote to equal the heating value.

We manufactured the E-SMAHC in two stages. First, we cured two anchoring tabs consisting of GFRP, KRAIBON[®], and the embedded SME wire (Figure 1A). We embedded the wire 20 mm into the tabs within the KRAIBON[®] interlayer, which proved sufficient to hold the wire until fracture without delamination. We tested other methods, but none were successful. Then we stretched the tabs to 5% pre-strain (Figure 1B). In the second stage, we incorporated this prestressed assembly into a new composite layup [GFRP $\pm 45^\circ_2$ / KRAIBON₁ / SMA / KRAIBON₁], co-curing Pt100 sensors, placed in contact with the SME wires within the laminate for direct temperature measurement (Figure 1C). The autoclave curing cycle was the same as for PO specimens.

To test the E-SMAHC, we clamped one end within a climatic chamber (Figure 1E). We ramped the temperature continuously while logging data from the two embedded Pt100 sensors and one ambient Pt100 sensor. A camera, positioned in front of the chamber window, recorded the cantilever's position at 10-second intervals, tracking three optical markers at 0, $L/2$, and L ($L = 81 \text{ mm}$, the E-SMAHC active length). We synchronised the temperature and curvature time series at $t = 0$, applied a Gaussian filter to suppress sensor noise without shifting the baseline and linearly interpolated the Pt100 signal at the optical timestamps to generate paired $\{T, \kappa\}$ data. Finally, we split this data at the peak temperature into heating and cooling branches.

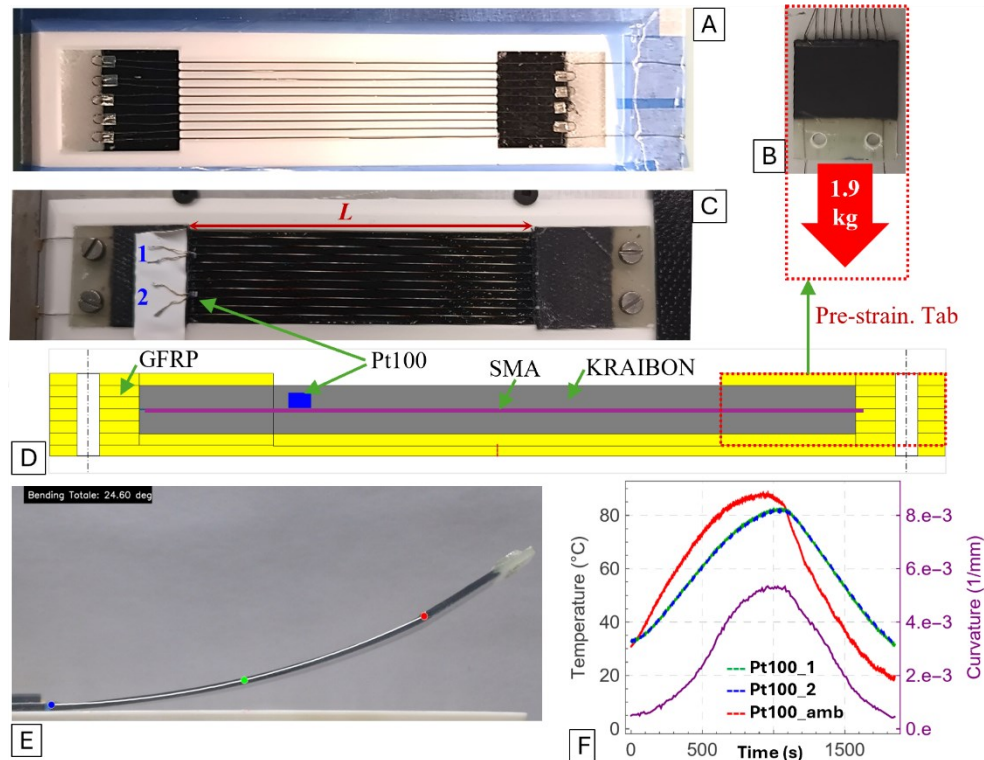


Figure 1. [A] Manufacturing of the pre-strain tabs with 3D printed fixture; [B] tensioning of the SME wire; [C] Manufacturing of the final E-SMAHC, fixing the pre-strain tabs and the 3D printed fixture, embedding Pt100 in contact with the SME wire; [D] layup of the E-SMAHC plies; [E] Elaborated image from the climatic chamber test with the three points and the curvature; [F] plot in time of the Pt100 temperatures and the curvature calculated from the three points.

3 Results and Discussion

3.1 Material and Characterisation

In tensile tests (Figure 2A), the wire showed an initial elastic modulus of 18 GPa and a detwinning plateau near 65 MPa at room temperature (*M*); under Joule heating (*A*), the modulus rose to 72 GPa with *SIM* onset around 650 MPa and UTS near 1300 MPa. For comparison, the SE wire in [2] had a Young's modulus of 65 GPa and a transformation plateau starting around 550 MPa. The constant-stress test (Figure 2B) confirmed the expected transformation temperatures.

Comparing KRAIBON® tensile tests between [2] and the present study (Figure 3B), we measured an initial linear Young's modulus of 8.3 MPa in the current batch, compared to 9.4 MPa previously, alongside a secondary slope of 0.95 MPa versus 1.32 MPa. We attribute this increased stiffness in the older product to natural ageing. We calculated the KRAIBON® hyperelastic fitting parameters identically to our previous methodology [2]: $D_1 = 0.0955 \text{ MPa}^{-1}$, and $C_{01}, C_{11}, C_{10}, C_{02}, C_{20} = 1.702, -0.369, -0.616, 0.429, 0.135 \text{ MPa}$ respectively.

3.2 Pull-Out Experiments

Figure 4 shows the PO force-displacement curves, the FEA results (for the no-pre-strain case only), and the behaviour of the free wire (between composite and clamp). Table 1 reports the maximum force and interfacial shear strength (IFSS) for each condition, along with results from [2] for comparison.

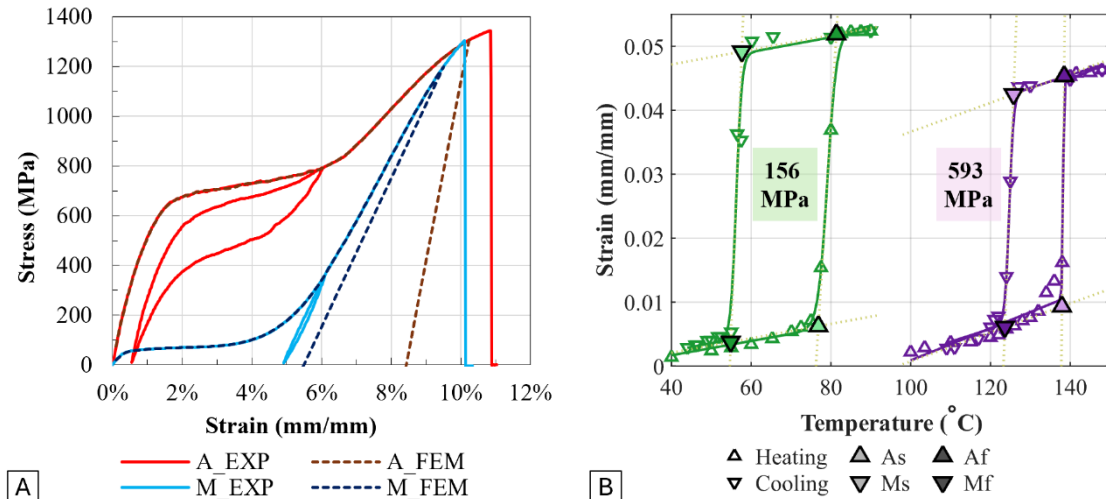


Figure 2. [A] tensile test at room temperature (*M*) and with joule-heating (*A*), and respective multilinear FEM; [B] constant stress testing for two different levels of stress and transformation temperature.

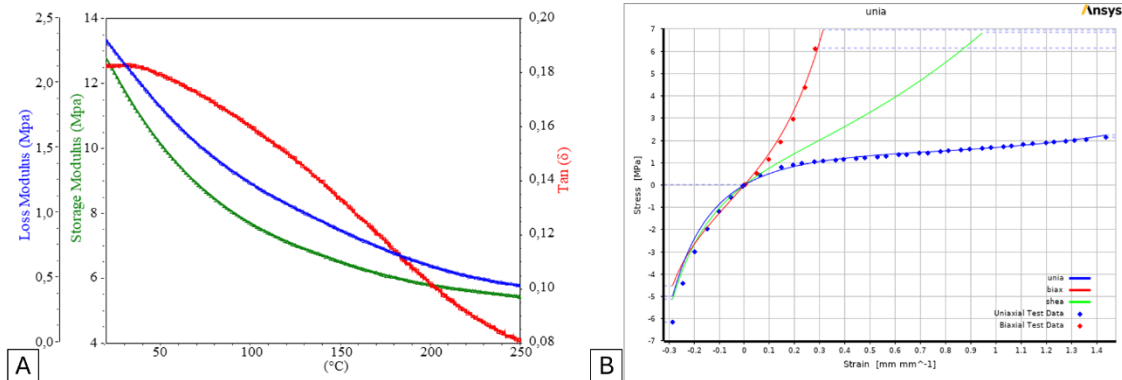


Figure 3. KRAIBON elastomer characterisation: [A] DMA storage and loss moduli, and $\tan \delta$, vs temperature; [B] Mooney–Rivlin fit on uniaxial tensile and (mirrored) compression data for the biaxial.

For 1/8" specimens, the pre-strained samples exhibited lower PO forces than the no-pre-strain samples, unexpectedly [5]. We attribute this anomaly to inadvertent tensioning introduced during the manual embedding process, an experimental sensitivity significantly amplified by the exceedingly short embedding length. The IFSS exhibited no significant difference between *M* and *A*, aligning with previous findings [2] (Tab. 1).

The 1" specimens behaved more consistently: all broke before the wire could pull out, showing that the KRAIBON® interface accommodates the wire transformation without debonding. Pre-strained specimens showed higher IFSS and stiffer initial response, consistent with [5]. Under Joule heating, the force needed to trigger the phase transformation was lower for the embedded wire than for the free wire. This reduction is likely attributable to heat dissipating into the matrix and the elastomer softening. The DMA in Figure 3A shows more than 30% reduction in stiffness between 25 °C and 90 °C. The same effect likely contributes to the FEA discrepancy.

The FEA reproduced force magnitudes within approximately 10% error for all specimen except 1" *M* with a 30% error and captured the overall shape of the force-displacement curve less accurately than in our previous models of SE wires. Because we deliberately retained the CZM parameters calibrated in our prior study, the current FEA functions primarily as a transferability validation rather than a new empirical fit.

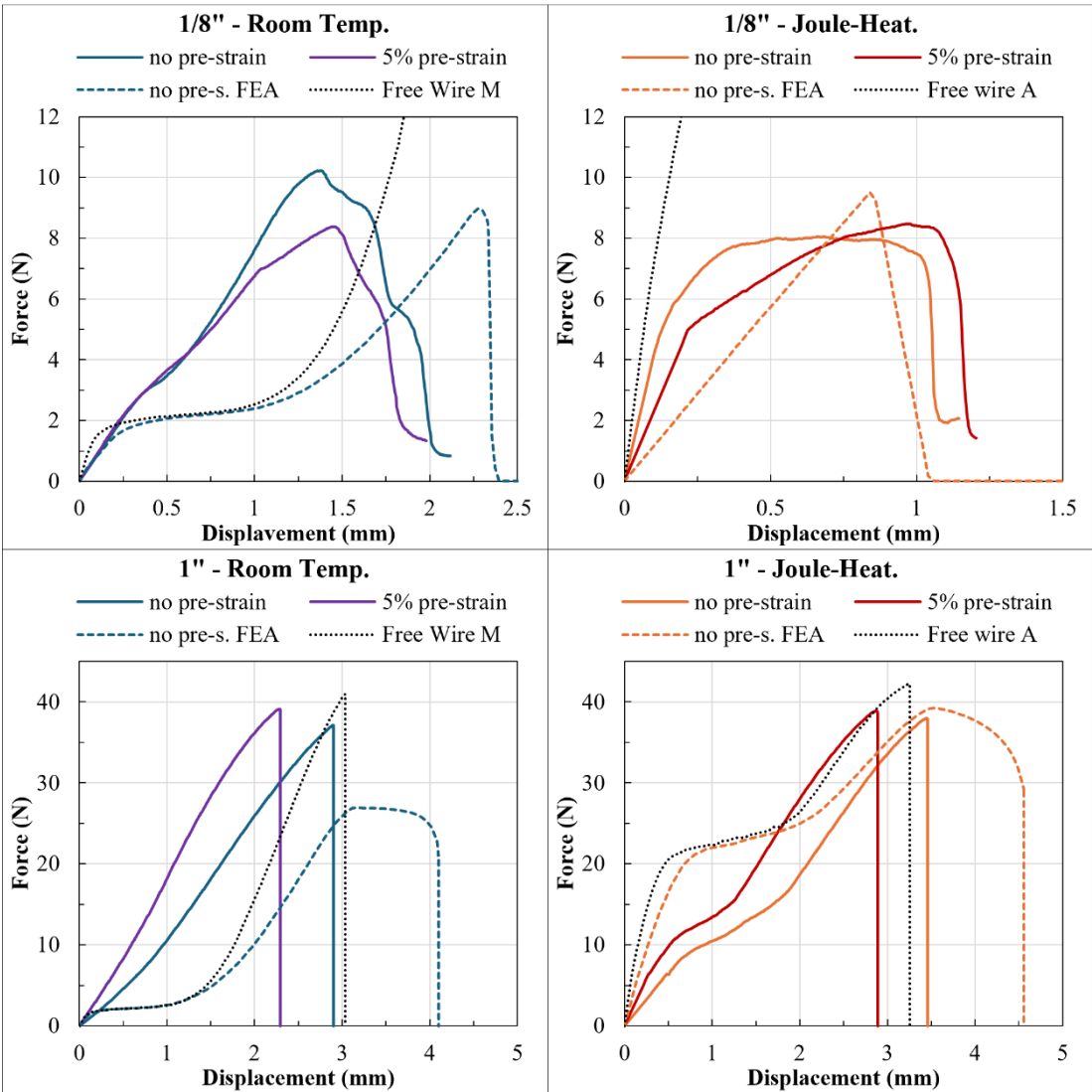


Figure 4. Pull-out test results for 2 embedding lengths, 1/8" and 1", in cool or heated state, with or without pre-strain; also FEA results for no pre-strain and the tensile test of the free wire are reported.

Table 1. Pull-out results: max force and interfacial shear stress from the present and previous study [2].

L_{embedded}	Phase	Pre-strain	Max Force [N]	IFSS [MPa]
1/8"	(M) Room T.	none	11.6 ± 1.0	5.7 ± 0.5
1/8"	(M) Room T.	5%	7.2 ± 0.9	3.6 ± 0.5
1/8"	(A) J.heated	none	6.6 ± 3.9	4.3 ± 0.3
1/8"	(A) J.heated	5%	7.1 ± 1.4	3.9 ± 0.4
1/8"	(A) SE [2]	none	9.0 ± 0.3	4.8 ± 0.1
1"	(M) Room T.	none	38 ± 3	2.4 ± 0.2
1"	(M) Room T.	5%	39 ± 2	2.5 ± 0.1
1"	(A) J.heated	none	37 ± 2	1.8 ± 0.1
1"	(A) J.heated	5%	38 ± 1	2.4 ± 0.1
1"	(A) SE [2]	none	28 ± 2	1.7 ± 0.1

3.3 E-SMAHC testing

Figure 1E shows the Pt100 temperatures and relative curvature, demonstrating good correspondence, as shown in Figure 5A as well. Figure 5B plots the discrete α_1 values calculated from Equation 2 for the embedded Pt100 sensor 1. The heating and cooling branches trace a hysteresis loop offset by approximately 10 °C. Table 2 reports the sigmoidal fit parameters for both heating and cooling branches.

Table 2. Sigmoidal fitting parameters as per Eq. 3 for the heating and the cooling branches.

	a ($10^{-3} / ^\circ\text{C}$)	b ($10^{-3} / ^\circ\text{C}$)	m ($^\circ\text{C}^{-1}$)	T_{mid} ($^\circ\text{C}$)	R^2
Heating	1.02	0.835	0.204	64.6	0.9999
Cooling	1.02	0.755	0.120	54.5	0.9995

Figure 5C compares the analytically predicted marker positions against the experimental measurements. Figure 5D shows the relative error, computed as the Euclidean distance between predicted and measured positions, normalised by the diagonal of the trajectory bounding box. The model achieved a tip RMSE of 1.15 mm (6.5% of full scale) and a mid-span RMSE of 0.6 mm (12% of full scale).

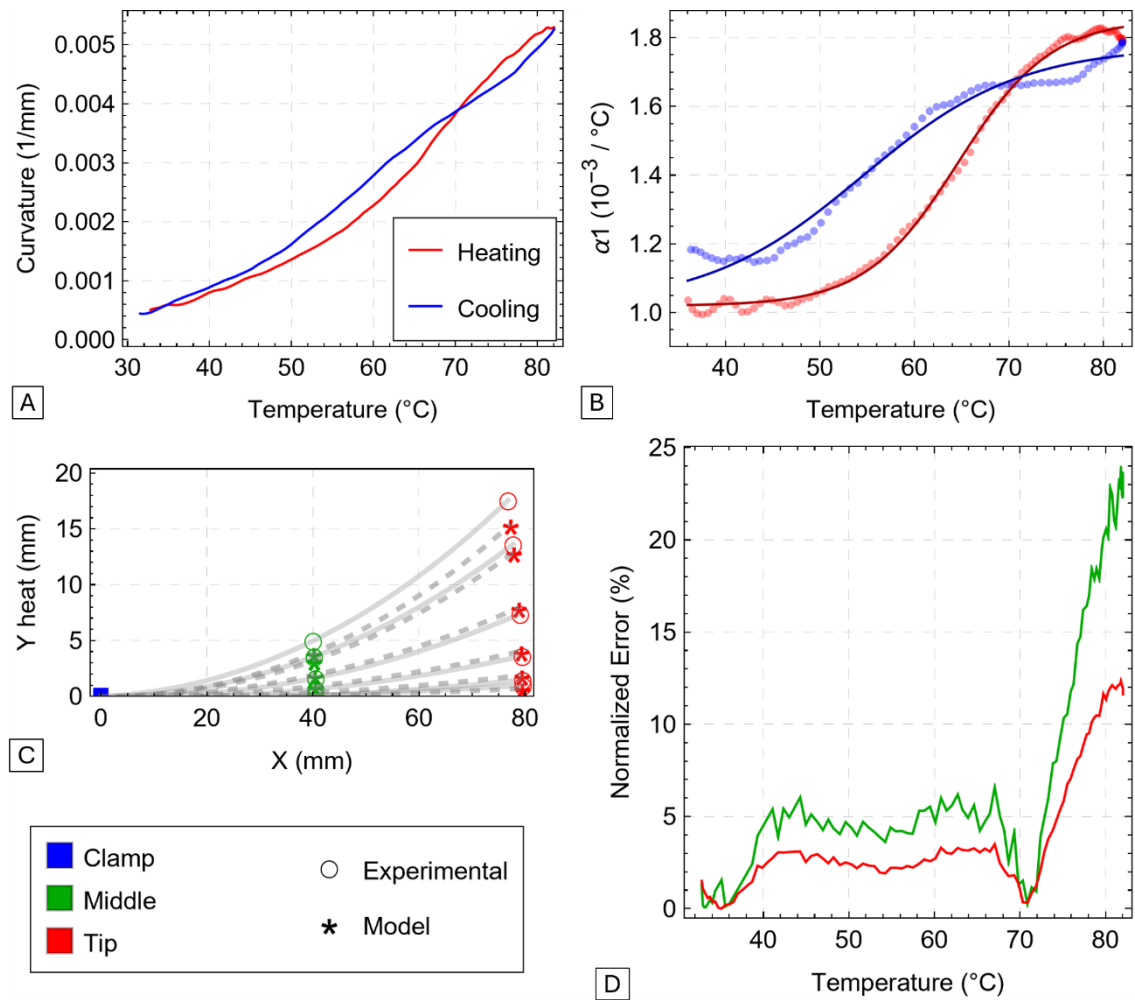


Figure 5. [A] curvature vs temperature heating and cooling branches; [B] discrete $\alpha_1(T)$ values plus sigmoidal fits; [C] computed vs experimental marker positions; [D] error normalised by maximum value.

Although the analytical model closely approximates the experimental results, certain limitations must be considered. First, the model is not inherently path-dependent, necessitating separate sigmoidal fits for the heating and cooling branches; capturing fast thermal transients would require integrating a dedicated hysteresis operator such as the Preisach model. Second, the current validation was performed without a tip load applied. Although we have formulated extended equations to incorporate tip loading, experimental validation remains future work. Third, the model relies on a one-dimensional assumption, presupposing uniform width and a symmetric distribution of wires. Fourth, the formulation is quasi-static, meaning it does not explicitly account for thermal transients or the viscoelastic time-dependence of the KRAIBON® interlayer. Finally, future work should benchmark this simplified model against higher-fidelity numerical approaches, such as classical laminate theory combined with Brinson constitutive laws [4] or fully coupled electro-thermo-mechanical finite element models [6].

Conclusions

All 1-inch KRAIBON-interfaced specimens incorporating SME wires reached wire fracture prior to debonding, across both martensitic and austenitic states, and at every pre-strain level evaluated. This outcome successfully extends the elastomeric interface concept from our previous work, originally demonstrated on superelastic wires, to the more demanding conditions of active shape-memory actuation.

Furthermore, our Timoshenko bimetallic beam model, employing a phenomenological CTE calibrated via sigmoidal fitting, accurately predicts cantilever tip deflection with an RMSE of 1.15 mm (6.5% of full scale) over the heating branch.

Finally, the implementation of a two-step autoclave protocol, supported by 3D-printed alignment masks and co-cured Pt100 sensors, yielded highly repeatable E-SMAHC cantilevers well-suited for future closed-loop control applications.

Future investigations will extend this validation to encompass tip loading, calibrate the exact current-to-temperature relationship for Joule heating, and characterise the full dynamic hysteresis loop necessary for precise closed-loop position control.

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